MATH 2470 - Homework 2

August 31, 2025

Problem 2.3. Three components (1, 2, and 3) are connected to form a system. Because the components in the 2–3 subsystem are connected in parallel, the whole systems works if component 1 function and one of component 2 and 3 works.

The experiment consists of determining the condition of each component: S (success) for a functioning component and F (failure) for a nonfunctioning component. For example, (S, S, S) is the outcome that all three components functions. (S, S, F) is the outcome that component 1 and 2 functions and component 3 does not work.

(a) Write out the event A that exactly two out of the three components function.

(b) Write out the event B that the system functions.

Hint: the solution should look like $A = \{$*(S,S,F), (, ,), . . .*$\}$

A = { (S,S,F), (S,F,S), (F,S,S) }

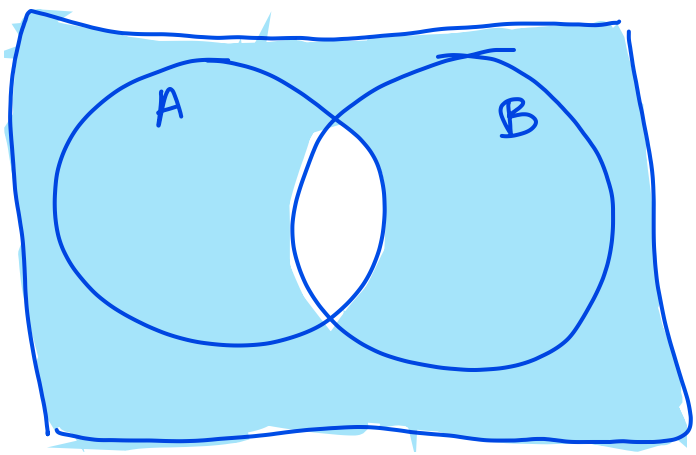
B = { (S,S,S), (S,S,F), (S,F,S) }

Problem 2.2. Use Venn diagrams to verify the following two relationships for any events A and B (these are called **De Morgan's laws**):

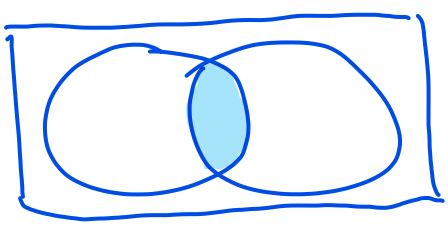
(a) $(A \cap B)' = A' \cup B'$

Hint: In each part, draw a diagram corresponding to the left side and another corresponding to the right side.

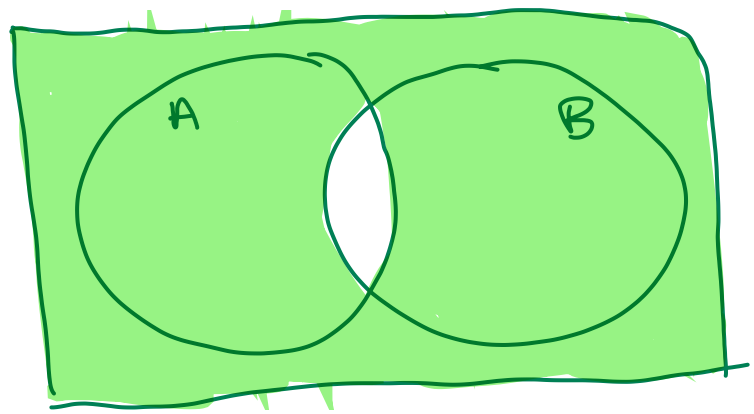
$(A \cap B)'$



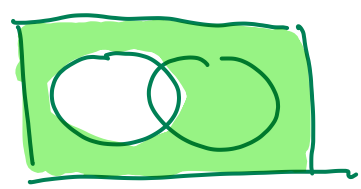
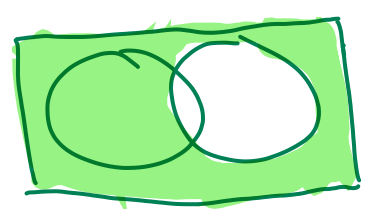
$A \cap B$



$A' \cup B'$



A'



B'